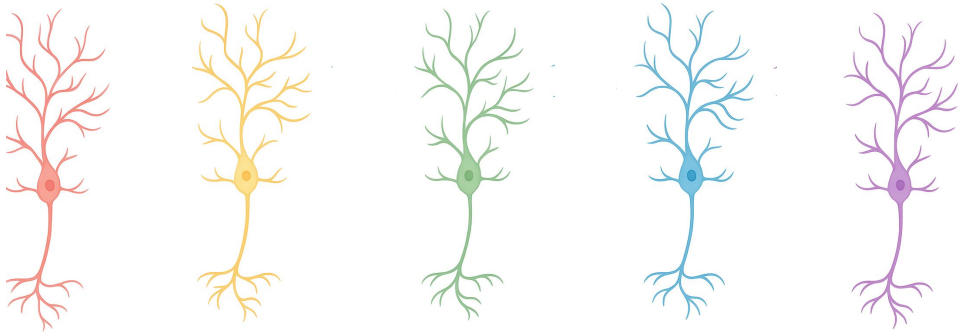
Prairie Voles

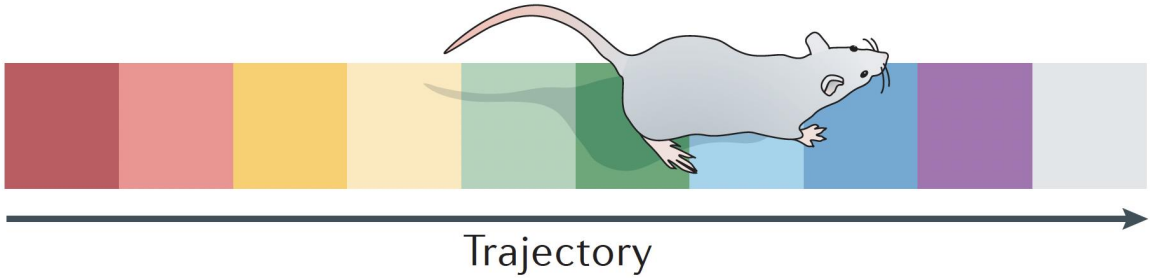
(and the evil code we wrote for them…)

Kelly, Autumn, Audrey

In 2014, John O'Keefe, May-Britt Moser, and Edvard Moser shared the Nobel Prize for their discoveries of place cells and grid cells

Hippocampal place cells





Cheese

Laura Colgin, *Nature Review Neuroscience*, 2016

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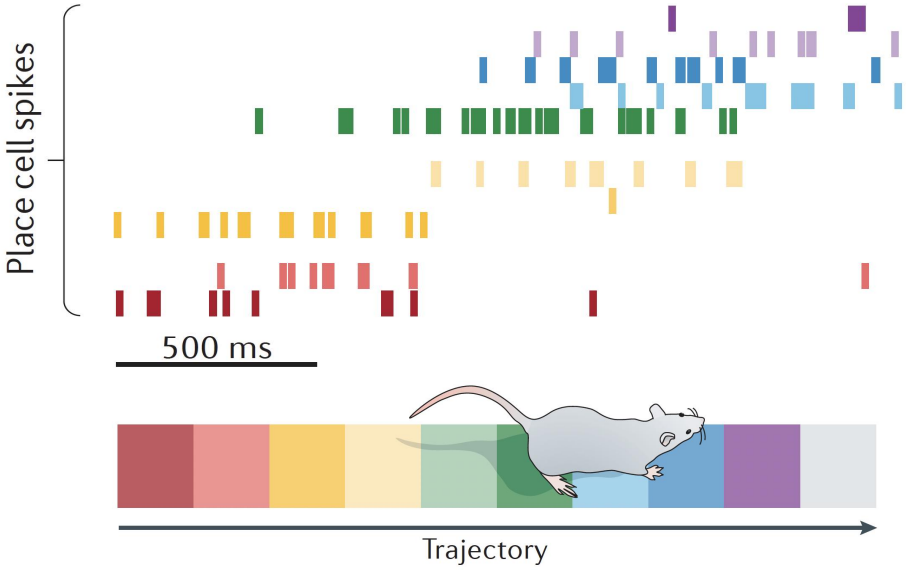


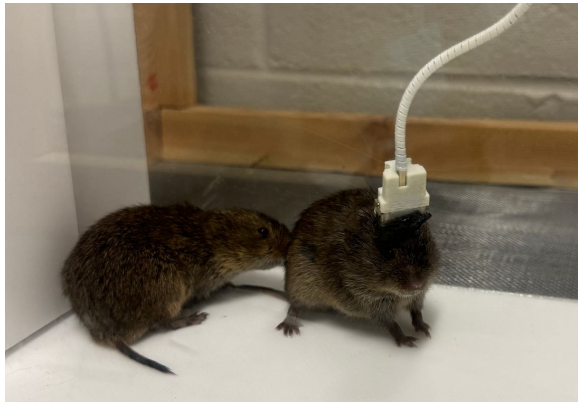
Active exploration

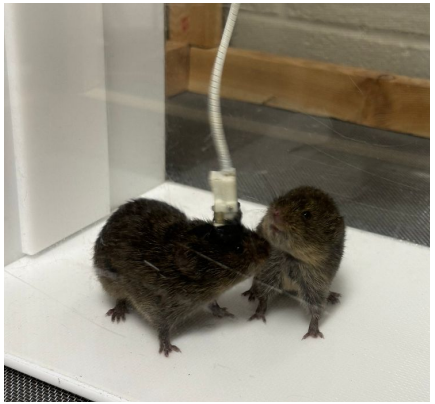
Making Memories

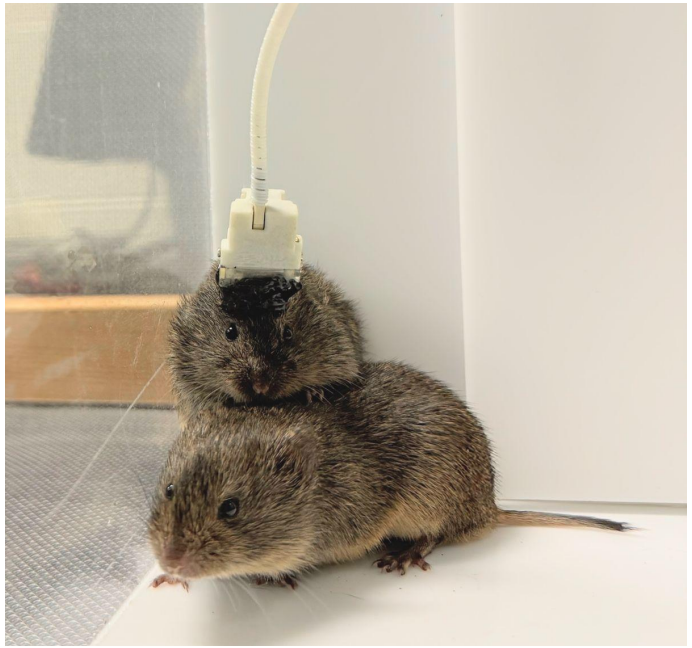
Sharp Wave Ripple

At rest

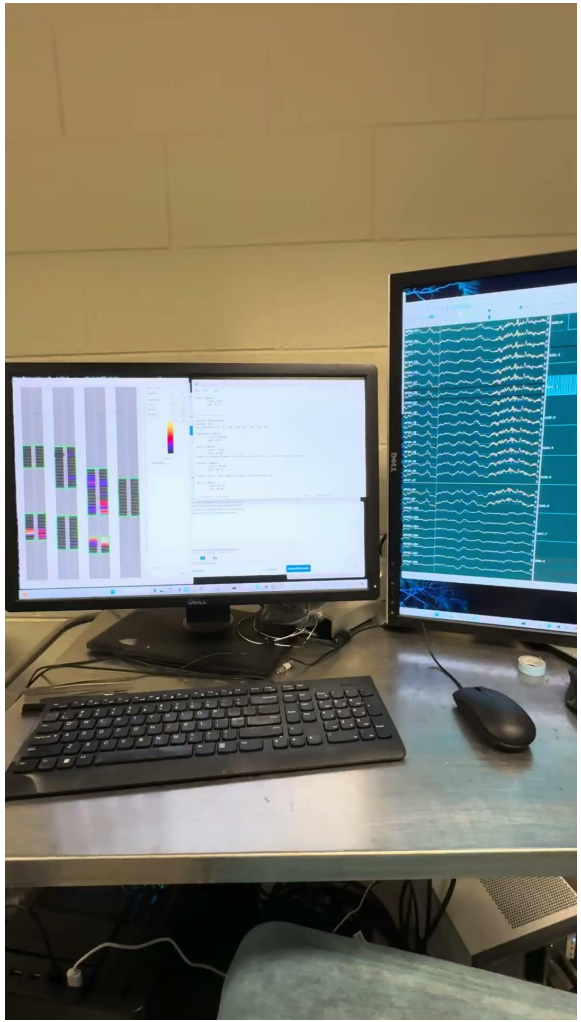
Is spiking activity sequential for social interactions?







get sniffed sniff cuddle



SWR.csv

start	peak	stop
100	150	300
450	520	550
…	…	…

behavior.csv

behavior	start	stop
social interaction	1	50
social interaction	60	70
sleeping	90	305
sleeping	445	560
…	…	…

spike_times.npy

1
1.01
1.02
1.03
…

cluster_ID.npy

2
4
23
5
…

cluster_KSLabel.tsv

1	good
2	good
3	mua
…	…

Step 1: Familiarize ourselves with the data products (provided by Kelly) and reading the data into a Python environment

Step 2: Re-organizing our data into accessible data structure(s) for easier access during analysis (i.e., dataframe, object, etc.)

Step 3: Write a function that will automatically search through data for co-activity during social interaction / sharp wave ripples

Step 4: Write visualization routines to help analyze neuron / cell assembly behavior

Step 5: Attempt to algorithmically predict / classify interactions based solely on neuron data

This was our initial project proposal…

…but how much did we *really* accomplish?

Step 1: Familiarize ourselves with the data products (provided by Kelly) and reading the data into a Python environment

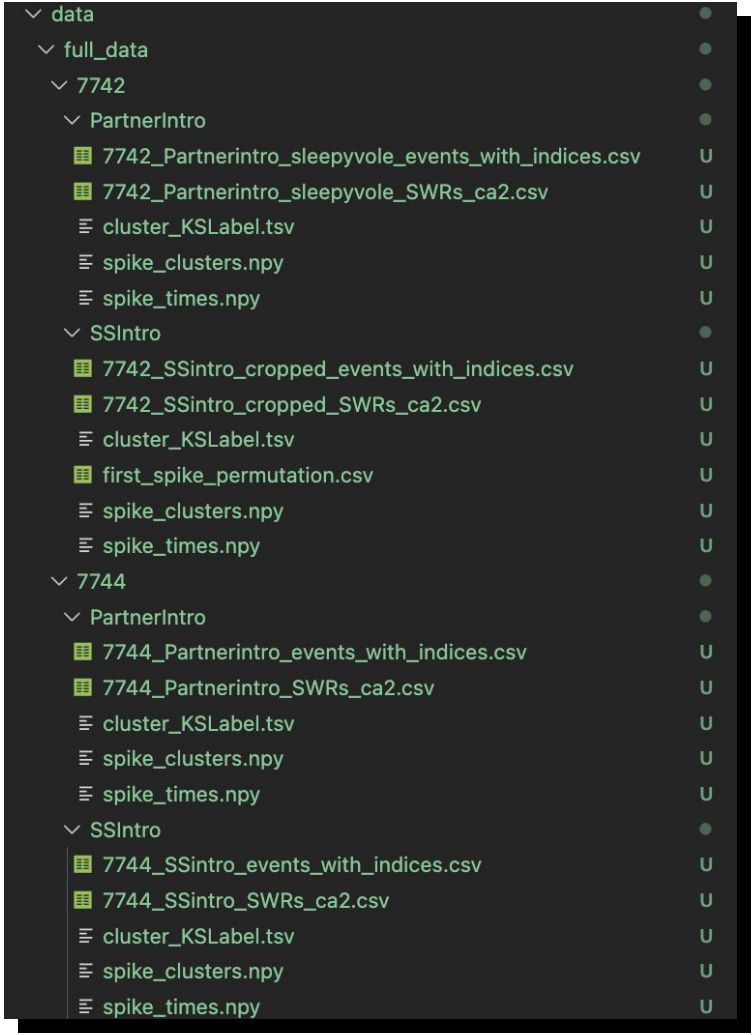
Step 2: Re-organizing our data into accessible data structure(s) for easier access during analysis (i.e., dataframe, object, etc.)

loading_utils.py

Is capable of…

- Loading data into Pandas DataFrames
- Merging datasets based on times
- Filtering DataFrames for specific values

  (example: good clusters only)



Step 3: Write a function that will automatically search through data for co-activity during social interaction / sharp wave ripples

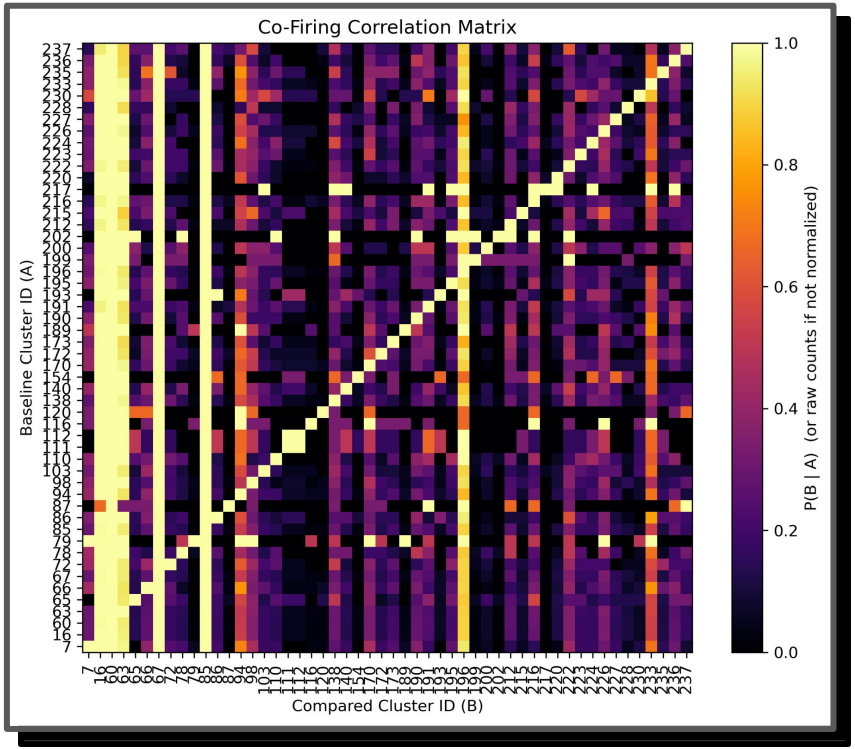
analysis_utils.py

sequence_matching_utils.py

Are capable of…

- Making matrices which represent the probability of Neuron B | Neuron A

“Unnormalized” Sequence Matches



Step 3: Write a function that will automatically search through data for co-activity during social interaction / sharp wave ripples

analysis_utils.py

sequence_matching_utils.py

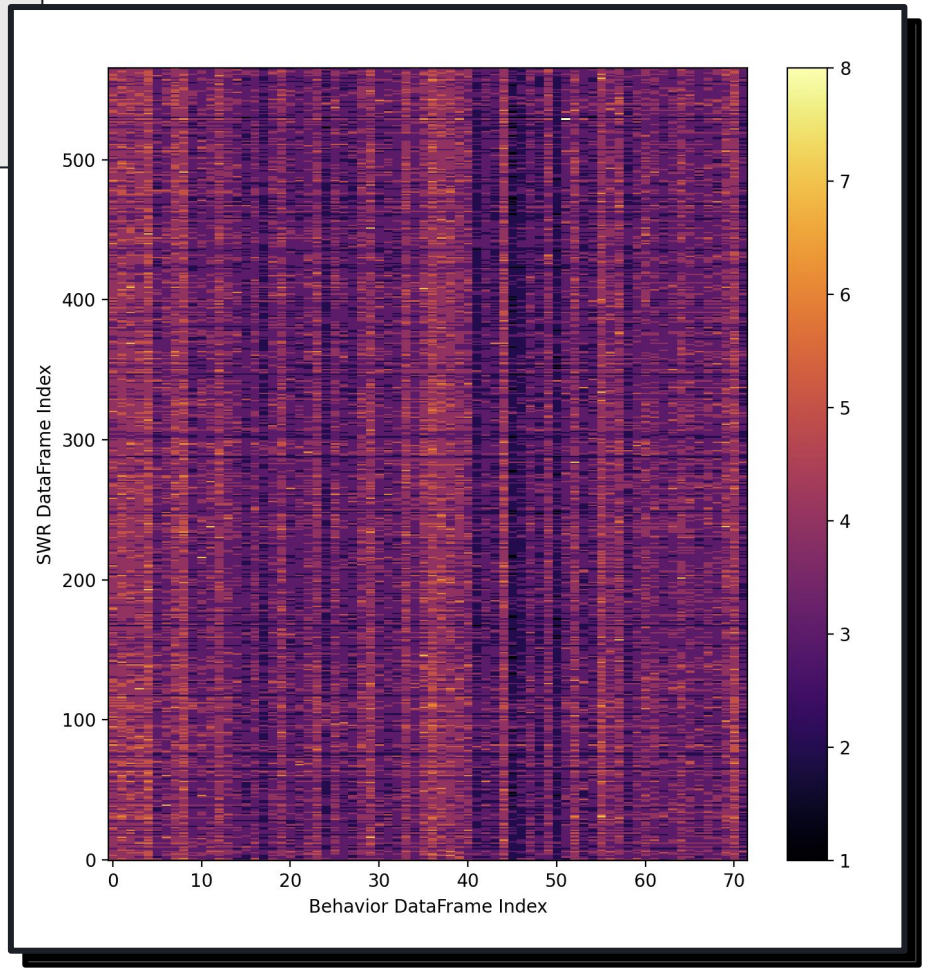
Are capable of…

- Making matrices which represent the probability of Neuron B | Neuron A
- Identifying sequences of neuron clusters that fire in two separate DataFrames

Future Direction…

- Save time of long matches and check video for clues of what might be replaying

“Unnormalized” Sequence Matches

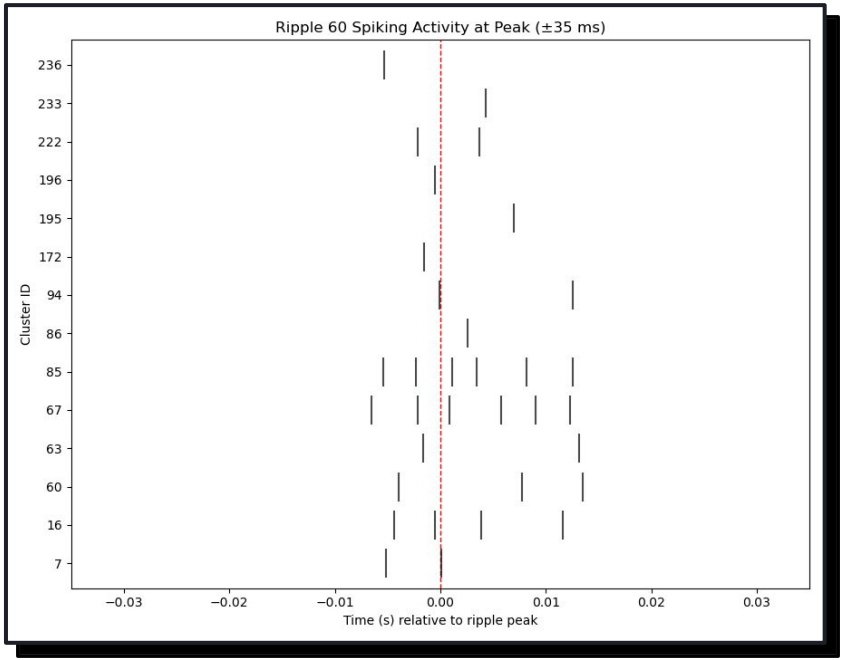


Step 4: Write visualization routines to help analyze neuron / cell assembly behavior

raster_plot_utils.py

Are capable of…

- Making raster plots for activity during individual SWRs or ‘Mega raster’ of all SWRs



Step 4: Write visualization routines to help analyze neuron / cell assembly behavior

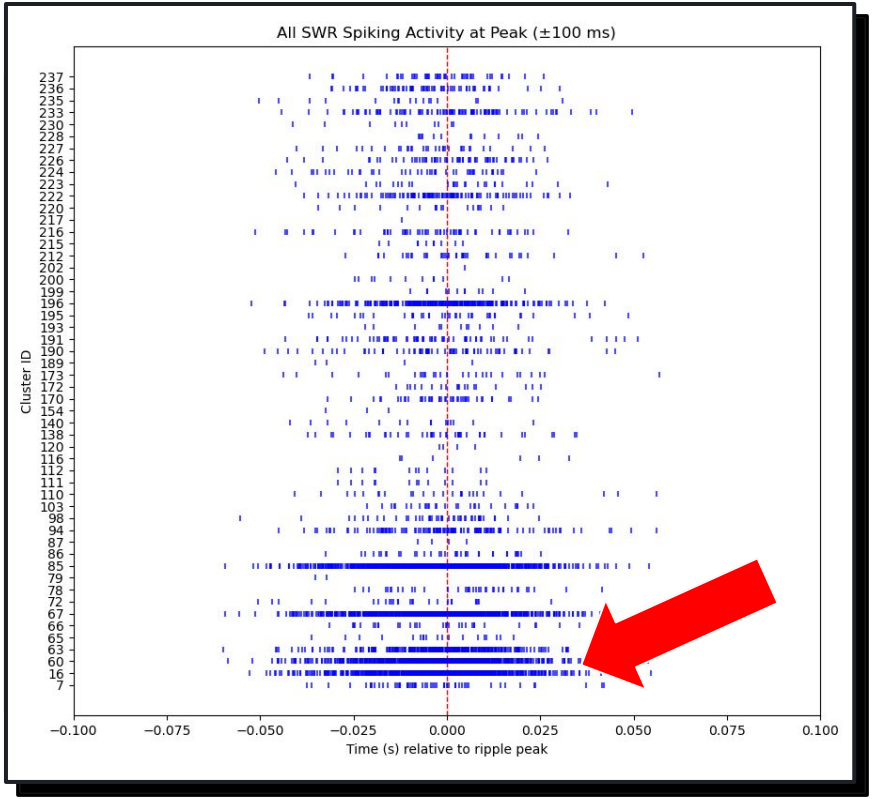
raster_plot_utils.py

Are capable of…

- Making raster plots for activity during individual SWRs or ‘Mega raster’ of all SWRs

WHAT IS THAT??

IS THAT NORMAL??



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raster_plot_utils.py

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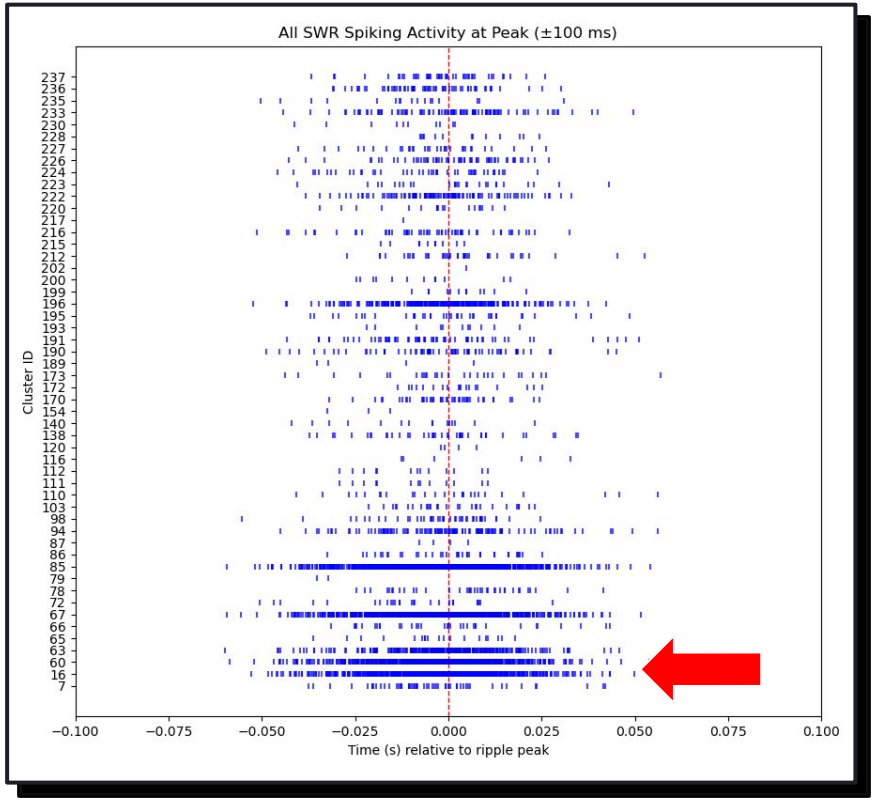


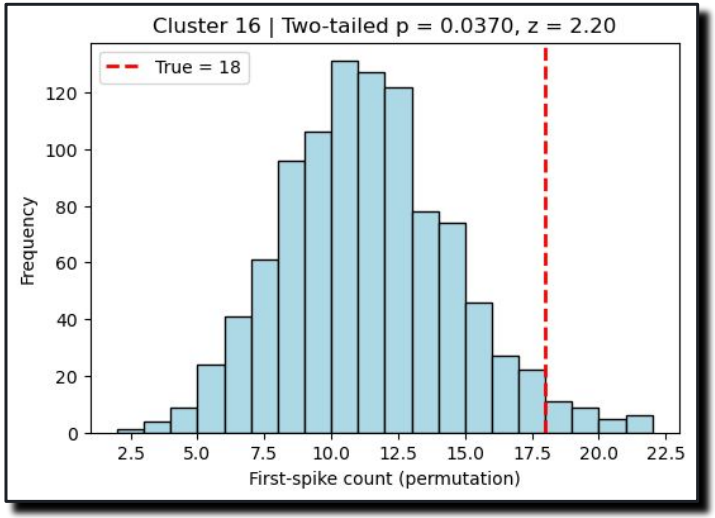
Step 4: Write visualization routines to help analyze neuron / cell assembly behavior

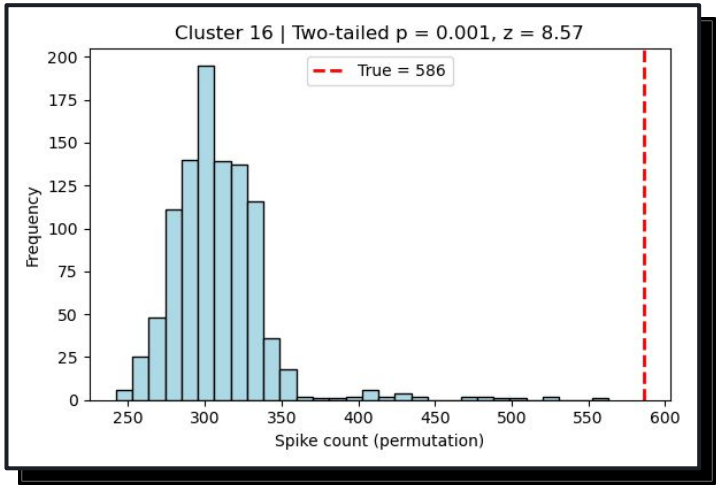
raster_plot_utils.py

analysis_utils.py

Are capable of…

- Making raster plots for activity during individual SWRs or ‘Mega raster’ of all SWRs
- Identify whether a neuron is spiking more than expected within a SWR
- In 1000 random shifts the SWR window to see if this is different than expected





Want to run it yourself?

Then *these files* are here to support you!

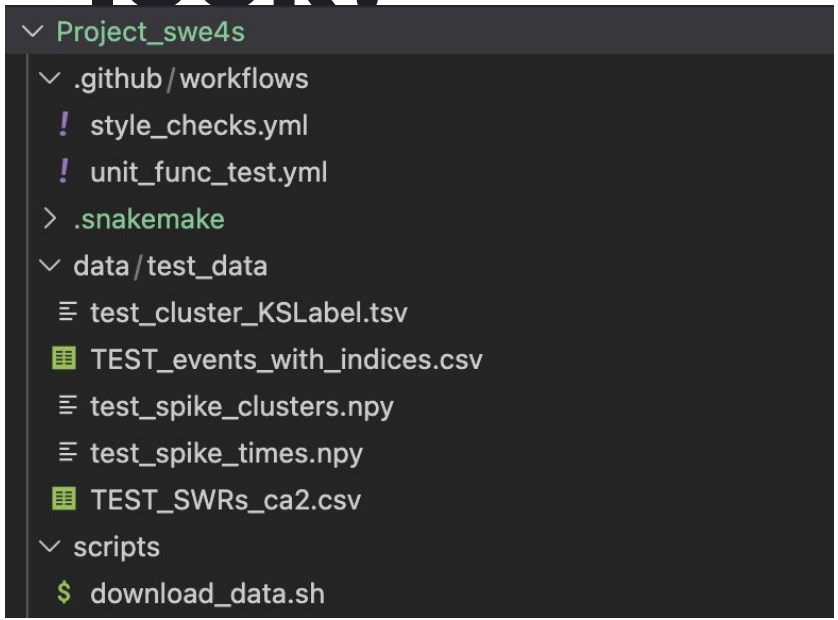
Future Directions?

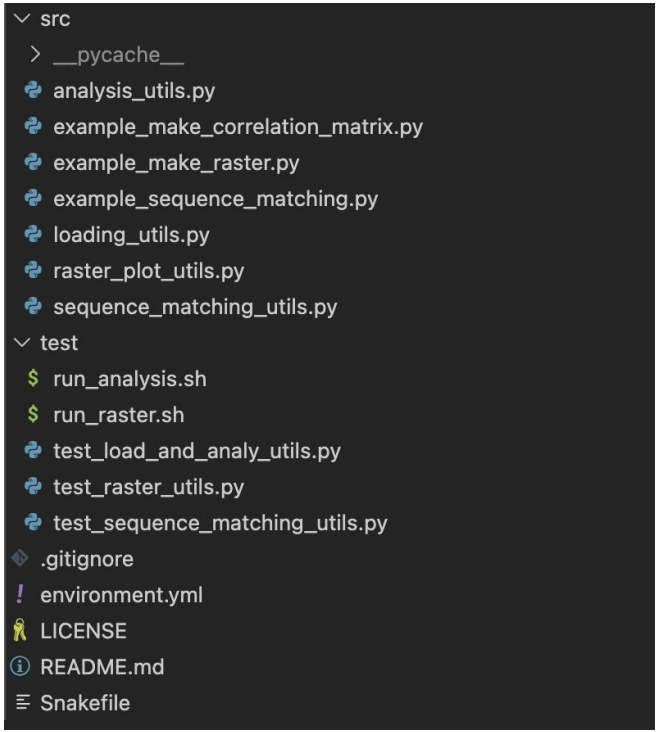
KELLY

It sure would be convenient to see if these functions could run any faster…

Also, I wonder if additional filtering for neuron types could provide unique insight into the sequence behavior!

How does the code look?





How are we testing it?

Unit Testing - Loading and Analysis Utils

test_load_and_analy_utils.py

- analysis_utils.py and loading_utils.py
- Responsible for testing:
  - Edge cases such as empty lists, missing columns and invalid modes
  - Circular permutation functional flow tests
  - Datatypes and output structures
  - Missing data errors
  - Bad file path errors

```
def test_match_times_empty_window(self):
    # Make SWR file with bad times
    fake_swr = pd.DataFrame({"Start": [99999], "Stop": [100000]})
    out = lu.group_dataframes_by_time(
        window_df=fake_swr,
        event_df=self.spike_df,
        event_time_column="Time",
        keep_event_columns=["Time"],
        time_interval_columns=["Start", "Stop"],
        progress=False
    )
    self.assertEqual(out.iloc[0]["Event Times"], [])
```

Unit Testing - Sequence Matching Utils

Test_sequence_match_utils.py

- Sequence_match_utils.py
- Responsible for Testing:
  - Correct handling of absent values
  - Correct computation
  - Hashing of lists and arrays
  - Normalization accuracy
  - Correct compute overlap matrix

```python
def test_prefix_hash(self):

    valid_array = [0, 4, 20, 200, 3]

    # Ensures that '_prefix_hash' works with lists and np.arrays
    self.assertTrue(
        utils._prefix_hash(valid_array, BASE, MOD),
        utils._prefix_hash(np.array(valid_array), BASE, MOD),
    )

    # Ensures that an empty list does return a hash (0)
    self.assertEqual(utils._prefix_hash([], BASE, MOD), [0])
```

Unit Testing - Raster Plots

test_raster_utils.py

- raster_plot_utils.py
- Responsible for testing:
  - Correct long form exploding
  - Correct t_rel calculation
  - Handling of string list inputs
  - Ripple filtering
  - Incorrect datatype handling
  - Invalid ripple index handling

```python
def test_select_all_ripples(self):
    # ripple_index=None should return the full exploded df
    out = rpu.select_ripples_to_plot(self.exp_df, ripple_index=None)
    self.assertEqual(len(out), len(self.exp_df))

def test_select_single_ripple(self):
    # Selecting a single ripple index returns only those rows
    out = rpu.select_ripples_to_plot(self.exp_df, ripple_index=0)
    self.assertEqual(set(out["ripple_idx"]), {0})
    # In test data ripple 0 has 3 spikes
    self.assertEqual(len(out), 3)

def test_select_multiple_ripples(self):
    # Multiple ripple indices returns a union of their rows
    out = rpu.select_ripples_to_plot(self.exp_df, ripple_index=[0, 1])
    self.assertEqual(set(out["ripple_idx"]), {0, 1})
    self.assertEqual(len(out), len(self.exp_df))
```

Functional Testing

```
valid_call ran in 5 sec with 2/16 lines to STDERR/OUT
Comparing output file to expected output file...
 PASS EXIT CODE (LINE 15)
```

run_analysis.sh

- Functional test for loading and analysis pipeline
- Uses example_make_correlation_matrix.py
  - Loads spike data
  - Match time (spike during a SWR)
  - Outputs correlation matrix
  - Visualize correlation matrix
- Checks output against expected data

run_raster.py

- Functional test for raster plot raster_plot_utils.py
- Uses example_make_raster.py
  - Load spike data
  - Match time (spike during a SWR)
  - Construct exploded raster dataframe
  - Apply ripple selection
  - Visualize raster plot
- Checks output against expected data